A Practical Hierarchical Reinforcement Learning Framework for Directional Slice-Aware Load Coordination and Mobility-Safe A3 Offset Control in O-RAN

Demand-Independent Global Reward with Directional Coordination

Corrected Two-Agent Formulation with Directional Global Bias and Continuous Local Control

Version 12.0 – Updated Reward Design

Abstract

This document presents a corrected two-agent hierarchical reinforcement learning (HRL) formulation for slice-aware load coordination and mobility-safe A3 offset control in O-RAN. The main design principle is the separation between strategic directional coordination and near-real-time mobility execution. The global agent, deployed as a Non-RT RIC rApp, observes network-wide slice load, SLA flags, and optionally connected-UE counters. It outputs a directional per-serving-gNB, per-neighbor, per-slice preference $b_{i,j,s} \in [-1, 1]$, indicating whether neighbor j is globally suitable for receiving traffic of slice s from serving gNB i.

The local agent, deployed as a near-RT RIC xApp, receives the global directional preference and fast local safety indicators such as SLA flags, previous offsets, handover failure ratio, and ping-pong ratio. It outputs continuous A3 proto-offsets $\widetilde{\mathrm{Offset}}_{i,j,s} \in [-6, +6]$ dB, which are then projected to the valid discrete A3 offset set before being applied by the gNB.

The updated contribution of this version is a demand-independent global reward. Since the controller is not responsible for user demand, the global reward no longer assumes that reducing total load is always possible or desirable. Instead, it rewards relative improvement in network-wide load distribution, saturation count, and SLA violations, while penalizing aggressive or unstable directional preferences. Therefore, the global agent learns to coordinate between gNBs under the current demand rather than trying to reduce the demand itself.

Contents

1 High-Level Idea

The objective is to coordinate slice load across neighboring gNBs without damaging mobility robustness. The corrected HRL design decomposes the problem into two different decisions:

1. **Strategic directional coordination:** identify which target-neighbor directions are globally suitable for offloading a given slice from a serving gNB.
2. **Tactical mobility execution:** decide the exact A3 offset to apply toward each neighbor and slice while preserving handover stability and SLA safety.

The corrected architecture is:

$$\begin{array}{c} \text{Global PPO agent in Non-RT RIC} \\ \Downarrow \quad \text{directional per-neighbor preference } b_{i,j,s} \\ \text{Local TD3 agent in near-RT RIC} \\ \Downarrow \quad \text{continuous proto-offset } \widetilde{\text{Offset}}_{i,j,s} \\ \text{Projection to valid A3 offset} \\ \Downarrow \quad \text{applied offset } \text{Offset}_{i,j,s}^{\text{applied}} \\ \text{gNB executes standard A3 handover rule.} \end{array}$$

The important correction is that the global action is directional:

$$b_{i,j,s}, \qquad j \in \mathcal{N}_i, \ \ s \in \mathcal{S}. \tag{1}$$

This means the global level does not only say that gNB i should offload slice s. It says whether offloading from i to each candidate neighbor j is globally useful. However, this is still not an A3 offset. The exact offset remains the responsibility of the local agent.

2 Why the Previous Design Needed Correction

The previous design used a global bias $b_{i,s}$. This bias was slice-aware, but it was not neighbor-aware. It could express:

$$b_{2,\text{eMBB}} < 0, \tag{2}$$

meaning that eMBB traffic at gNB2 should be offloaded. However, it could not express whether gNB2 should offload toward gNB1 or gNB3.

This created a logical weakness. The local agent had to select the target neighbor, but the local agent was intentionally denied PRB load values. Therefore, it could choose a neighbor with stable mobility but poor load capacity. Connected-UE counters are not enough to solve this issue because UE count is not equivalent to PRB load. A small number of high-rate eMBB users may consume more PRBs than many low-rate mMTC devices.

The corrected design solves this by making the global action directional:

$$b_{i,j,s} \in [-1, 1]. \tag{3}$$

The global agent uses network-wide load and SLA information to judge whether neighbor j is globally suitable for receiving slice s traffic from gNB i. The local agent then uses near-real-time mobility indicators to decide how aggressive the A3 offset should be.

3 System Model

3.1 Network, Slices, and Neighbors

Let

$$\mathcal{G} = \{1, 2, \ldots, N\} \tag{4}$$

denote the set of gNBs. Let

$$\mathcal{S} = \{\mathrm{URLLC}, \mathrm{eMBB}, \mathrm{mMTC}\} \tag{5}$$

denote the set of slices. The formulation also works for any number $S = |\mathcal{S}|$ of slices.

For each serving gNB $i \in \mathcal{G}$, let

$$\mathcal{N}_i \subset \mathcal{G} \setminus \{i\} \tag{6}$$

be the set of candidate target neighbors. The maximum number of neighbors is denoted by $d_{\max}$. In implementation, variable neighbor counts can be handled using padding and action masks, or using graph/attention-based neighbor encoders.

3.2 Slice Load

For every gNB i and slice s, the normalized PRB load is

$$L_{i,s}(t) = \frac{\mathrm{PRB}_{\mathrm{used},i,s}(t)}{\mathrm{PRB}_{\mathrm{total},i,s}(t)} \in [0, 1]. \tag{7}$$

This is the main load coordination measurement. It is observed by the global agent, not by the local agent.

3.3 Connected UE Counter

For each gNB and slice, the number of connected UEs is

$$K_{i,s}(t) = \sum_{u \in \mathcal{U}_i(t)} \mathbf{1}[\mathrm{slice}(u) = s]. \tag{8}$$

The connected UE counter may be observed by the global agent. It may also be used by the local agent only as an optional lightweight context variable, but not as a replacement for PRB load.

3.4 Slice-Specific SLA Constraints

Each slice has a different service requirement:

$$\begin{cases} D_{i,\mathrm{URLLC}}(t) \le D_{\max}, \\ R_{i,\mathrm{eMBB}}(t) \ge R_{\min}, \\ Q_{i,\mathrm{mMTC}}(t) \le Q_{\max}. \end{cases} \tag{9}$$

These are converted into constraint functions:

$$c_{i,\mathrm{eMBB}}(t) = R_{\min} - R_{i,\mathrm{eMBB}}(t), \tag{10}$$

$$c_{i,\mathrm{URLLC}}(t) = D_{i,\mathrm{URLLC}}(t) - D_{\max}, \tag{11}$$

$$c_{i,\mathrm{mMTC}}(t) = Q_{i,\mathrm{mMTC}}(t) - Q_{\max}. \tag{12}$$

A violation occurs when $c_{i,s}(t) > 0$, and the violation flag is

$$v_{i,s}(t) = \mathbf{1}[c_{i,s}(t) > 0]. \tag{13}$$

3.5 Mobility Counters

For each serving gNB i, target neighbor j, and slice s, define:

- $N_{i,j,s}^{\text{HOA}}$: number of handover attempts,
- $N_{i,j,s}^{\text{HOS}}$: number of successful handovers,
- $N_{i,j,s}^{\text{HOF}}$: number of handover failures,
- $N_{i,j,s}^{\text{PP}}$: number of ping-pong handovers.

The mobility ratios are

$$R_{i,j,s}^{\text{HF}}(t) = \frac{N_{i,j,s}^{\text{HOF}}(t)}{N_{i,j,s}^{\text{HOA}}(t) + \epsilon}, \tag{14}$$

$$R_{i,j,s}^{\text{PP}}(t) = \frac{N_{i,j,s}^{\text{PP}}(t)}{N_{i,j,s}^{\text{HOS}}(t) + \epsilon}, \tag{15}$$

where $\epsilon > 0$ avoids division by zero.

3.6 A3 Handover Model

The simplified A3 condition is written using RSRP:

$$\text{RSRP}_j(t) > \text{RSRP}_i(t) + \text{Offset}_{i,j,s}^{\text{applied}}(t). \tag{16}$$

A negative offset makes handover easier toward neighbor j, while a positive offset makes handover harder. The valid offset set is

$$\mathcal{A}_{\text{offset}} = \{-6, -4, -2, 0, +2, +4, +6\} \text{ dB}. \tag{17}$$

The learning system does not directly select individual UE handovers. It only adapts slice-aware neighbor-specific A3 offsets. The gNB still executes the standard A3 rule.

4 Corrected Two-Agent HRL Formulation

4.1 Role Separation

The corrected design is based on the following separation:

Component	Question answered	Main information
Global PPO rApp	Which neighbor directions are globally useful for slice offloading?	Network-wide load, SLA flags, optional UE counters
Local TD3 xApp	How aggressive should the A3 offset be right now?	Directional preference, handover failures, ping-pong, previous offset, SLA flags
gNB	Which UEs satisfy the A3 event?	UE measurements such as RSRP

The global agent is load-aware and neighbor-aware, but it is not an offset controller. The local agent is mobility-aware and offset-aware, but it is not a second load-balancing controller.

4.2 Timescales

Agent	**Location**	**Period**
Global agent	Non-RT RIC rApp	$T_H = 5$ s
Local agent	near-RT RIC xApp	$T_L = 500$ ms

The global agent updates slowly because load coordination decisions should be stable. The local agent updates faster because mobility conditions and handover stability can change over short timescales.

5 Global Agent Formulation

5.1 Role

The global agent answers:

> Which target neighbor directions are globally suitable for offloading slice s from serving gNB i?

It does not output the final A3 offset. It outputs a directional preference.

5.2 Global State

The basic global state is

$$s_H(t) = [L_{i,s}(t)]_{i,s} \parallel [v_{i,s}(t)]_{i,s}. \tag{18}$$

Optionally, connected UE counters can be included:

$$s_H(t) = [L_{i,s}(t)]_{i,s} \parallel [v_{i,s}(t)]_{i,s} \parallel [K_{i,s}(t)]_{i,s}. \tag{19}$$

The global state dimension is $2NS$ without UE counters and $3NS$ with UE counters.

5.3 Directional Global Action

The corrected global action is

$$a_H(t) = B(t) = [b_{i,j,s}(t)]_{i\in\mathcal{G},j\in\mathcal{N}_i,s\in\mathcal{S}}. \tag{20}$$

Each entry satisfies

$$b_{i,j,s}(t) \in [-1, 1]. \tag{21}$$

The interpretation is:

- $b_{i,j,s} < 0$: offloading slice s from gNB i toward neighbor j is globally encouraged;
- $b_{i,j,s} \approx 0$: neutral;
- $b_{i,j,s} > 0$: offloading slice s from gNB i toward neighbor j is discouraged.

This action is directional but not operational. It does not directly configure the A3 offset. The local agent still converts it into a safe offset.

For a fixed maximum neighbor count $d_{\max}$, the global action dimension is

$$\dim(a_H) = N d_{\max} S. \tag{22}$$

If the neighbor graph is sparse, an action mask is used so that invalid neighbor entries do not affect the environment or the policy loss.

5.4 Demand-Independent Global Reward

The global reward must account for an important practical constraint: the controller is not responsible for the traffic demand generated by users. Demand depends on UE activity, requested throughput, radio conditions, slice reservation, and whether UEs are actually movable. Therefore, the reward should not simply penalize high total load or reward lower total load. The global PPO agent should instead be evaluated by whether its directional preferences improve the network configuration under the current demand.

The reward is formulated using a global network cost $J_H(t)$ and an action regularization term $J_{\text{action}}(t)$:

$$r_H(t) = J_H(t) - J_H(t + T_H) - J_{\text{action}}(t) - J_{\text{bad\_dir}}(t). \tag{23}$$

A positive reward means that the global network state after one global control period is better than before, after accounting for unnecessary or unsafe global actions.

5.4.1 Global Network Cost

A simple demand-independent network cost is

$$J_H(t) = \mu_H \sum_{s \in \mathcal{S}} \text{Var}_{i \in \mathcal{G}}(L_{i,s}(t)) + \zeta_H N_{\text{sat}}(t) + \beta_H N_{\text{SLA}}(t). \tag{24}$$

The first term measures imbalance across gNBs for each slice. This does not penalize total demand directly; it only penalizes poor distribution. The second term counts saturated gNB/slice pairs. The third term counts SLA violations.

The saturation count is defined as

$$N_{\text{sat}}(t) = \sum_{i,s} \mathbf{1}[L_{i,s}(t) \geq L_s^{\text{crit}}], \tag{25}$$

where L_s^{crit} is a slice-specific critical load threshold, for example 0.95. The SLA violation count is

$$N_{\text{SLA}}(t) = \sum_{i,s} v_{i,s}(t). \tag{26}$$

If continuous SLA violation magnitudes are available, $N_{\text{SLA}}(t)$ can be replaced by

$$\sum_{i,s} \max(0, c_{i,s}(t)). \tag{27}$$

5.4.2 Action Regularization

The global action regularizer discourages unnecessary aggressive offloading and unstable global decisions:

$$J_{\text{action}}(t) = \lambda_H \sum_{i,j,s} \max(0, -b_{i,j,s}(t))^2 + \kappa_H \sum_{i,j,s} (b_{i,j,s}(t) - b_{i,j,s}(t - T_H))^2 . \tag{28}$$

The first term penalizes strong negative preferences because negative $b_{i,j,s}$ values encourage offloading. The second term penalizes sudden changes in the global directional preference, making the global coordination smoother and more stable.

5.4.3 Bad-Direction Penalty

To make inter-gNB coordination more explicit, the reward includes a bad-direction penalty:

$$J_{\text{bad\_dir}}(t) = \eta_H \sum_{i,j,s} \max(0, -b_{i,j,s}(t)) \max(0, L_{j,s}(t) - L_{i,s}(t)). \tag{29}$$

This term penalizes the global agent when it encourages offloading from i to j while the target neighbor j is already more loaded than the serving cell i for the same slice. It does not forbid such actions completely, but it requires them to be justified by the overall reward.

A target-side SLA version can also be added:

$$J_{\text{unsafe\_target}}(t) = \rho_H \sum_{i,j,s} \max(0, -b_{i,j,s}(t)) v_{j,s}(t), \tag{30}$$

which discourages offloading toward target cells whose SLA is already violated.

5.4.4 Recommended Final Reward

The recommended global reward is

$$\begin{aligned}
r_H(t) = {} & J_H(t) - J_H(t + T_H) && (31)\\
& - \lambda_H \sum_{i,j,s} \max(0, -b_{i,j,s}(t))^2 && (32)\\
& - \kappa_H \sum_{i,j,s} (b_{i,j,s}(t) - b_{i,j,s}(t - T_H))^2 && (33)\\
& - \eta_H \sum_{i,j,s} \max(0, -b_{i,j,s}(t)) \max(0, L_{j,s}(t) - L_{i,s}(t)). && (34)
\end{aligned}$$

with

$$J_H(t) = \mu_H \sum_s \mathrm{Var}_i(L_{i,s}(t)) + \zeta_H N_{\text{sat}}(t) + \beta_H N_{\text{SLA}}(t). \tag{35}$$

This reward makes the global agent coordinate between gNBs because the reward is computed from the whole network and the action is directional. The agent is rewarded only if its directional preferences improve the global load distribution, reduce saturation, or reduce SLA violations under the same demand conditions.

5.4.5 No-Action Baseline Variant

In a simulator, an even cleaner demand-independent reward can be computed by comparing the agent to a neutral-action baseline. Let $J_H^{\text{agent}}(t + T_H)$ be the cost after applying the global action $B(t)$, and let $J_H^{\text{neutral}}(t + T_H)$ be the cost after applying a neutral action $b_{i,j,s} = 0$ under the same traffic and mobility realization. Then

$$r_H(t) = J_H^{\text{neutral}}(t + T_H) - J_H^{\text{agent}}(t + T_H) - J_{\text{action}}(t) - J_{\text{bad\_dir}}(t). \tag{36}$$

This version is the most scientifically clean because both rollouts experience the same exogenous demand. The reward becomes positive only when the global agent improves the outcome compared with doing nothing.

5.4.6 Suggested Initial Weights

For a first simulation campaign, the following weights are recommended:

Weight	Suggested value
μ_H	1.0
ζ_H	1.0
β_H	5.0
λ_H	0.01
κ_H	0.05
η_H	0.5
ρ_H	1.0 if target-side SLA penalty is used

The weights should be normalized after measuring the empirical range of each term in simulation. The action penalties should remain weaker than the global state improvement terms, otherwise the agent may become too conservative and output only neutral preferences.

5.5 Global Policy

The global policy is

$$\pi_H : \mathbb{R}^{2NS} \to [-1, 1]^{Nd_{\max}S}. \tag{37}$$

A typical neural network is:

$$\text{Input} \to \text{FC}(256) \to \text{ReLU} \to \text{FC}(128) \to \text{ReLU} \to \text{FC}(Nd_{\max}S) \to \tanh.$$

PPO is used because the global action is continuous and the upper-level decision operates at a slow timescale.

5.6 Global Training Algorithm

1. Initialize global policy π_H and value network V.
2. Freeze or deploy the current local TD3 executor.
3. At each global step, observe network-wide state $s_H(t)$.
4. Output directional preference matrix $B(t) = \pi_H(s_H(t))$.
5. Send each $b_{i,j,s}(t)$ to the local executor responsible for serving gNB i.
6. Simulate T_H seconds while local agents execute faster A3 offset control.
7. Compute the demand-independent global reward from the improvement in J_H, action regularization, and bad-direction penalties.
8. Update π_H using PPO and generalized advantage estimation.

6 Local Agent Formulation

6.1 Role

The local agent answers:

> Given the global directional preference and near-real-time mobility safety indicators, what A3 offset should be applied toward neighbor j for slice s?

The local agent does not decide whether a neighbor is globally load-feasible. That has already been compressed into $b_{i,j,s}$ by the global agent.

6.2 Local State

For each neighbor $j \in \mathcal{N}_i$ and slice $s \in \mathcal{S}$, the corrected local observation block is

$$o_{i,j,s}(t) = \Big[b_{i,j,s}(t),\ v_{i,s}(t),\ v_{j,s}(t),\ \mathrm{Offset}^{\mathrm{prev}}_{i,j,s}(t),\ R^{\mathrm{HF}}_{i,j,s}(t),\ R^{\mathrm{PP}}_{i,j,s}(t)\Big]. \tag{38}$$

Optional lightweight counters can be added:

$$K_{i,s}(t), \quad K_{j,s}(t), \tag{39}$$

only if they are interpreted as context signals, not as PRB load substitutes. The cleanest non-redundant version excludes direct PRB load and can also exclude neighbor UE counts.

The full local observation is

$$o_i(t) = \bigoplus_{s\in\mathcal{S}} \bigoplus_{j\in\mathcal{N}_i} o_{i,j,s}(t). \tag{40}$$

With six features per neighbor-slice block, the dimension is

$$\dim(o_i) = 6Sd_i. \tag{41}$$

For shared local TD3 with variable neighbor counts, use $d_{\max}$ with masks:

$$\dim(o_i) = 6Sd_{\max}. \tag{42}$$

6.3 Continuous Local Action

The local actor outputs one continuous proto-offset for every neighbor-slice pair:

$$\widetilde{a}_{L,i}(t) = \Big[\widetilde{\mathrm{Offset}}_{i,j,s}(t) : j \in \mathcal{N}_i, s \in \mathcal{S}\Big] \in [-6, +6]^{Sd_i}. \tag{43}$$

The actor network can be written as

$$o_i \to \mathrm{FC}(256) \to \mathrm{ReLU} \to \mathrm{FC}(128) \to \mathrm{ReLU} \to \mathrm{FC}(Sd_{\max}) \to 6\tanh(\cdot).$$

6.4 Projection to Valid A3 Offsets

The gNB receives only a valid A3 offset. Therefore,

$$\mathrm{Offset}^{\mathrm{applied}}_{i,j,s}(t) = Q\left(\widetilde{\mathrm{Offset}}_{i,j,s}(t)\right), \tag{44}$$

where

$$Q(x) = \arg\min_{a\in\mathcal{A}_{\mathrm{offset}}} |x - a|. \tag{45}$$

Ties can be rounded toward the value with smaller magnitude to avoid unnecessarily aggressive changes.

6.5 Local Reward

The local reward should not contain a direct load-balancing term based on $L_{i,s}$ or $L_{j,s}$ because the local level must not duplicate the global load coordination role. Instead, the local reward focuses on safe execution of the global directional preference.

A practical local reward is:

$$r_{L,i}(t) = -\alpha_L \sum_{j,s} R^{\mathrm{HF}}_{i,j,s}(t) - \beta_L \sum_{j,s} R^{\mathrm{PP}}_{i,j,s}(t) \tag{46}$$

$$- \gamma_L \sum_{s} \left[\max(0, c_{i,s}(t)) + \omega_T \max(0, c_{j,s}(t))\right] \tag{47}$$

$$- \delta_L \sum_{j,s} \left(\widetilde{\mathrm{Offset}}_{i,j,s}(t) - \mathrm{Offset}^{\mathrm{prev}}_{i,j,s}(t)\right)^2 \tag{48}$$

$$+ \eta_L \sum_{j,s} b_{i,j,s}(t)\widetilde{\mathrm{Offset}}_{i,j,s}(t). \tag{49}$$

The first two terms penalize handover failure and ping-pong behavior. The third term penalizes SLA violations at both the serving cell and the target cell. The target-cell SLA term is important because an offloading action from i to j can improve the serving cell while harming the target cell. The fourth term discourages sudden offset oscillations.

The final term is a small alignment reward. Under the chosen convention, $b_{i,j,s} < 0$ means that offloading toward neighbor j is encouraged, and a negative A3 offset makes handover toward j easier. Therefore, when both $b_{i,j,s}$ and $\widetilde{\mathrm{Offset}}_{i,j,s}$ are negative, their product is positive and the alignment reward increases. Similarly, when $b_{i,j,s} > 0$, a positive or neutral offset is encouraged. This term should be weaker than the SLA and mobility penalties so that the local agent can reduce aggressiveness when the recommended direction becomes unsafe.

6.6 Lagrangian SLA Penalty

A Lagrangian version can be used for SLA constraints:

$$\lambda_{i,s} \leftarrow \max(0, \lambda_{i,s} + \eta_\lambda \bar{c}_{i,s}), \tag{50}$$

where

$$\bar{c}_{i,s} = \frac{1}{T_{\mathrm{ep}}} \sum_{t} \max(0, c_{i,s}(t)). \tag{51}$$

For local actions that affect a target neighbor, a target-side multiplier can also be used:

$$\lambda_{j,s} \max(0, c_{j,s}(t)). \tag{52}$$

6.7 TD3 Actor-Critic Structure

The local agent uses TD3 because the local action is continuous. Each local executor contains:

- an actor $\mu_\theta(o_i)$ that outputs proto-offsets;
- two critics $Q_{\phi_1}(o_i, \widetilde{a}_{L,i})$ and $Q_{\phi_2}(o_i, \widetilde{a}_{L,i})$.

The two critics reduce overestimation by using the minimum target value.

7 Step-by-Step Scenario with Three gNBs

7.1 Initial Situation

Consider three gNBs:

$$\mathcal{G} = \{\mathrm{gNB1}, \mathrm{gNB2}, \mathrm{gNB3}\} \tag{53}$$

and one slice for illustration:

$$s = \mathrm{eMBB}. \tag{54}$$

The current normalized loads are:

gNB	eMBB load
gNB1	0.50
gNB2	0.92
gNB3	0.65

Assume gNB2 has two neighbors:

$$\mathcal{N}_2 = \{\text{gNB1}, \text{gNB3}\}. \tag{55}$$

Mobility indicators are:

Direction	Handover failure ratio	Ping-pong ratio
gNB2 → gNB1	0.08	0.04
gNB2 → gNB3	0.25	0.30

7.2 Global PPO Output

The global agent sees network-wide load and SLA information. It outputs directional preferences:

$$b_{2,1,\text{eMBB}} = -0.9, \tag{56}$$

$$b_{2,3,\text{eMBB}} = +0.4. \tag{57}$$

This means that offloading eMBB from gNB2 toward gNB1 is strongly encouraged, while offloading eMBB from gNB2 toward gNB3 is discouraged. The global agent has not selected an A3 offset. It has only expressed directional coordination intent.

7.3 Local TD3 Observation

The local executor at gNB2 receives, for the direction gNB2 → gNB1:

$$o_{2,1,\text{eMBB}} = [-0.9, v_{2,\text{eMBB}}, v_{1,\text{eMBB}}, 0, 0.08, 0.04]. \tag{58}$$

For the direction gNB2 → gNB3:

$$o_{2,3,\text{eMBB}} = [+0.4, v_{2,\text{eMBB}}, v_{3,\text{eMBB}}, 0, 0.25, 0.30]. \tag{59}$$

The local executor sees the global preference and the mobility risk. It does not need to observe the PRB load of gNB1 or gNB3 because the global agent already used that information.

7.4 Local TD3 Output and Quantization

The local TD3 actor outputs:

$$\widetilde{\text{Offset}}_{2,1,\text{eMBB}} = -4.2, \tag{60}$$

$$\widetilde{\text{Offset}}_{2,3,\text{eMBB}} = +2.7. \tag{61}$$

After quantization:

$$\text{Offset}^{\text{applied}}_{2,1,\text{eMBB}} = -4 \text{ dB}, \tag{62}$$

$$\text{Offset}^{\text{applied}}_{2,3,\text{eMBB}} = +2 \text{ dB}. \tag{63}$$

Thus, handover toward gNB1 becomes easier, while handover toward gNB3 becomes harder.

7.5 A3 Execution

For gNB2 $\to$ gNB1:

$$\text{RSRP}_1 > \text{RSRP}_2 - 4. \tag{64}$$

For gNB2 $\to$ gNB3:

$$\text{RSRP}_3 > \text{RSRP}_2 + 2. \tag{65}$$

Therefore, UEs near the gNB2/gNB1 boundary are more likely to satisfy the A3 condition toward gNB1, while handovers toward the risky gNB3 direction are reduced.

7.6 Result

Suppose five eMBB UEs successfully move from gNB2 to gNB1. The loads become:

gNB	Old load	New load
gNB1	0.50	0.62
gNB2	0.92	0.77
gNB3	0.65	0.65

The hierarchy works because the global agent selected gNB1 as the better load coordination target, the local agent selected the offset intensity according to mobility risk, and the gNB executed standard A3 handover without direct UE-level RL control.

7.7 Reward Interpretation in This Scenario

The global reward does not assume that the total eMBB demand decreases. It evaluates whether the post-action distribution is better than before. If gNB2 becomes less overloaded but gNB1 becomes saturated, N_{sat} may increase and the reward becomes poor. If gNB2 improves and gNB1 remains safe, $J_H(t + T_H)$ decreases and the global PPO receives a positive reward.

7.8 Why the Lower Agent Still Matters

Now suppose the global agent still sees gNB1 as a good target, but the near-real-time mobility counters change:

Direction	Handover failure ratio	Ping-pong ratio
gNB2 $\to$ gNB1	0.35	0.40

The global preference may remain

$$b_{2,1,\text{eMBB}} = -0.9, \tag{66}$$

because gNB1 still has spare load capacity. However, the local TD3 can reduce the aggressiveness:

$$\widetilde{\text{Offset}}_{2,1,\text{eMBB}} = -0.8, \tag{67}$$

which quantizes to 0 or -2 dB depending on the chosen quantization rule. This shows why the local agent is still necessary. The global agent decides that the direction is useful from a load perspective, but the local agent decides whether the offset should be aggressive or conservative under current mobility conditions.

8 Training Workflow

8.1 Recommended Three-Phase Strategy

The corrected training workflow is:

Phase 1: Local TD3 pretraining with rule-based directional preferences. The global PPO is not trained yet. Directional preferences are generated by a simple heuristic to expose the local agent to offload, neutral, and retain commands.

Phase 2: Global PPO training with frozen local TD3. The local policy is frozen and deployed at all gNBs. The global PPO learns which directional preferences improve network-wide coordination under demand-independent reward feedback.

Phase 3: Joint fine-tuning. Both levels are fine-tuned with smaller learning rates while preserving the meaning of $b_{i,j,s}$.

8.2 Curriculum Integration

The traffic and mobility curriculum should be applied inside the training phases, not before them as an unrelated sequence. The recommended curriculum is:

Stage A: Static UEs near handover boundaries with variable demand. This verifies that offset changes can actually move users. Static UEs far from handover boundaries should not be the main training case because A3 offsets may not produce useful handovers.

Stage B: Static UEs with arrival and departure. This tests adaptation to changes in user population and demand.

Stage C: Slow controlled mobility. This introduces natural A3 handovers and tests mobility penalties.

Stage D: SUMO-based mobility. This is the final realistic training and evaluation stage.

8.3 Local Pretraining Rule

For pretraining only, a simple rule can generate directional global preferences:

$$b_{i,j,s}^{\text{rule}} = \begin{cases} -1, & L_{i,s} > 0.75 \text{ and } L_{j,s} < 0.60, \\ +1, & L_{j,s} > 0.80 \text{ or } v_{j,s} = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{68}$$

This rule is not the final controller. It is only used to pretrain the local TD3 agent to understand the semantics of directional preferences.

8.4 Global PPO Training

After local pretraining, the local TD3 is frozen. PPO then learns $b_{i,j,s}$ from reward. This is what makes the global part reinforcement learning rather than fixed if-else logic. The semantic interpretation of the action is defined, but the mapping from state to action is learned.

The updated reward is especially suitable for this phase because it evaluates improvement in the whole network while avoiding the unrealistic assumption that the global agent can reduce user demand.

8.5 Joint Fine-Tuning

During joint fine-tuning, both agents are updated with smaller learning rates. The local learning rate should be especially small so that the meaning of $b_{i,j,s}$ remains stable. Otherwise, the global agent may learn that a given bias value produces one offset behavior, while the local policy later changes that behavior.

9 Validation

9.1 Simulator Sanity Checks

Before validating learning, the simulator must satisfy:

- negative offsets increase handover attempts toward the target neighbor;
- positive offsets reduce handover attempts toward the target neighbor;
- successful handovers update $K_{i,s}$, $K_{j,s}$, $L_{i,s}$, and $L_{j,s}$ consistently;
- handover failures and ping-pong events are counted over a sliding window;
- the reward does not penalize the agent directly for exogenous demand increases.

9.2 Local TD3 Validation

The local TD3 is valid if it learns to convert a directional preference into a safe offset. For example:

- if $b_{i,j,s} < 0$ and mobility is stable, the local output should be negative;
- if $b_{i,j,s} < 0$ but handover failure or ping-pong is high, the local output should be less aggressive;
- if $b_{i,j,s} > 0$, the local output should be neutral or positive.

Metrics include local reward, handover failure ratio, ping-pong ratio, SLA violation rate, offset oscillation magnitude, and sign alignment between $b_{i,j,s}$ and $\widetilde{\text{Offset}}_{i,j,s}$.

9.3 Global PPO Validation

The global PPO is valid if it learns to reduce load imbalance, reduce saturation count, and avoid SLA degradation without assuming control over demand. Metrics include:

- slice-load variance;
- saturation count N_{sat};
- global SLA violation count N_{SLA};
- bad-direction penalty magnitude;
- consistency of directional preferences;
- average global reward;
- performance compared with a neutral-action baseline.

9.4 Complete HRL Validation

The full HRL model is successful if

$$\mathrm{Var}(L)\downarrow,\quad N_{\mathrm{sat}}\downarrow,\quad N_{\mathrm{SLA}}\downarrow, \tag{69}$$

while

$$R^{\mathrm{HF}} \text{ remains low},\quad R^{\mathrm{PP}} \text{ remains low},\quad \text{HO count remains controlled.} \tag{70}$$

Validation episodes should run without exploration noise and should use unseen traffic distributions, UE placements, mobility seeds, and SUMO routes.

10 Baselines

The final evaluation should compare:

1. fixed default A3 offset;
2. neutral global action $b_{i,j,s} = 0$ with local TD3;
3. rule-based directional preference with rule-based local offset;
4. rule-based directional preference with local TD3;
5. global PPO directional preference with rule-based local offset;
6. full HRL: global PPO directional preference with shared local TD3 offset execution.

The neutral global action baseline is especially important for validating the demand-independent reward, because it separates the effect of the controller from exogenous traffic demand.

11 Complexity Analysis

11.1 Global Complexity

With directional preferences, the global action dimension is

$$O(Nd_{\max}S). \tag{71}$$

With fixed hidden layers, one global forward pass scales as

$$O(Nd_{\max}S). \tag{72}$$

11.2 Local Complexity

For one local agent, the local action dimension is

$$O(d_iS). \tag{73}$$

For N local executors, the forward-pass complexity is

$$O(NSd_{\max}). \tag{74}$$

A naive flat discrete joint-action formulation would require enumerating

$$7^{Sd_i} \tag{75}$$

offset combinations per local agent. This is only the complexity of the naive flat discrete formulation; factorized discrete policies may reduce this cost, but the continuous TD3 formulation avoids such enumeration directly.

12 O-RAN Integration

12.1 Component Mapping

O-RAN component	Role
Non-RT RIC rApp	Hosts the global PPO agent and computes directional preferences $b_{i,j,s}$ every T_H.
A1 interface	Sends directional policies or preference vectors to the near-RT RIC.
near-RT RIC xApp	Hosts local TD3 executors and computes proto-offsets every T_L.
E2SM-KPM	Reports load and SLA values to the global rApp; reports mobility counters and SLA flags to the local xApp.
E2SM-RC	Sends quantized neighbor-specific slice A3 offsets to the gNB, assuming the RAN exposes the required slice-aware control hooks.
O-CU-CP	Applies A3 offset configuration and executes handover control.
O-DU	Provides radio measurements such as RSRP.

12.2 Deployment Assumption

This formulation assumes that the RAN exposes slice-aware or slice-conditioned mobility control parameters, or that vendor-specific/programmable control allows slice-specific A3 offset policies. If the deployed RAN only supports cell-level or neighbor-level A3 offsets without slice conditioning, the action space must be adapted accordingly.

12.3 Example Control Message


```
{
  "action_type": "A3_OFFSET_CONFIG",
  "target_gNB": "gNB-2",
  "neighbor_slice_configs": [
    {
      "target_neighbor": "gNB-1",
      "slice": "eMBB",
      "global_preference": -0.90,
      "proto_offset_dB": -4.20,
      "applied_offset_dB": -4
    },
    {
      "target_neighbor": "gNB-3",
      "slice": "eMBB",
      "global_preference": 0.40,
      "proto_offset_dB": 2.70,
      "applied_offset_dB": 2
    }
  ],
  "validity_duration_ms": 500
}
```

13 Implementation Notes

- Expose the global action as a flattened vector of length $Nd_{\max}S$ with masks for invalid neighbors.
- Reshape the global vector into directional preferences $b_{i,j,s}$ before sending to local executors.
- Do not give direct PRB load $L_{i,s}$ or $L_{j,s}$ to the local agent.
- Keep connected-UE counters optional in the local state; if used, clearly state that they are context counters, not load measurements.
- Store continuous proto-actions in the TD3 replay buffer.
- Execute quantized offsets in the simulator.
- Compute mobility KPIs over sliding windows to reduce noise.
- Use action masks for missing neighbors when sharing local TD3 parameters across gNBs.
- Evaluate validation episodes without exploration noise.
- Compute the global reward from relative improvement, not from absolute total demand.
- For the cleanest demand-independent evaluation, compare the agent rollout with a neutral-action rollout under the same traffic and mobility seed.

14 Conclusion

This corrected formulation keeps the HRL architecture meaningful by separating directional load-aware coordination from near-real-time mobility-safe offset execution. The global PPO agent observes network-wide slice load and SLA conditions and outputs directional preferences $b_{i,j,s}$. These preferences identify which neighbor directions are globally suitable for offloading, but they are not A3 offsets. The local TD3 agent receives these preferences along with mobility safety indicators and outputs continuous proto-offsets that are projected to valid A3 values before application.

The updated demand-independent reward improves the formulation by acknowledging that the controller is not responsible for user demand. Instead of rewarding total load reduction, the global reward evaluates whether the action improves network-wide distribution, reduces saturation, and avoids SLA degradation under the current demand. The action regularizer and bad-direction penalty encourage conservative, stable, and coordinated inter-gNB behavior. Therefore, the hierarchy remains defensible: global PPO decides where offloading is useful, local TD3 decides how safely and aggressively to execute it, and the gNB performs the actual UE-level handover using the standard A3 rule.

A Acronyms

Acronym	Meaning
3GPP	3rd Generation Partnership Project
A1	Interface between Non-RT RIC and near-RT RIC
A3	Handover event type
CMDP	Constrained Markov Decision Process
E2	Interface between near-RT RIC and O-DU/O-CU
eMBB	enhanced Mobile Broadband
gNB	Next Generation Node B
HRL	Hierarchical Reinforcement Learning
mMTC	massive Machine Type Communications
Non-RT RIC	Non-Real-Time RAN Intelligent Controller
near-RT RIC	Near-Real-Time RAN Intelligent Controller
PPO	Proximal Policy Optimization
PRB	Physical Resource Block
RSRP	Reference Signal Received Power
SLA	Service-Level Agreement
TD3	Twin Delayed Deep Deterministic Policy Gradient
URLLC	Ultra-Reliable Low-Latency Communications
xApp	RIC Application for near-RT RIC

B References

[1] 3GPP TS 38.331, "Radio Resource Control (RRC) protocol specification," 2022.

[2] O-RAN Alliance, "O-RAN Architecture Description," Technical Specification, 2023.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," arXiv:1707.06347, 2017.

[4] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing Function Approximation Error in Actor-Critic Methods," ICML, 2018.

[5] E. Altman, "Constrained Markov Decision Processes," Chapman & Hall/CRC, 1999.